
INET Framework Developer's Guide

Release 4.6.0

Feb 27, 2026

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Release: 4.6.0

INTRODUCTION

1.1 What is INET Framework

The INET Framework is an open-source model library for the OMNeT++ simulation environment. It provides protocols, agents, and other models for researchers and students working with communication networks. INET is especially useful when designing and validating new protocols or exploring new or exotic scenarios.

INET supports a wide class of communication networks, including wired, wireless, mobile, ad hoc, and sensor networks. It contains models for the Internet stack (TCP, UDP, IPv4, IPv6, OSPF, BGP, etc.), link-layer protocols (Ethernet, PPP, IEEE 802.11, various sensor MAC protocols, etc), refined support for the wireless physical layer, MANET routing protocols, DiffServ, MPLS with LDP and RSVP-TE signalling, several application models, and many other protocols and components. It also provides support for node mobility, advanced visualization, network emulation, and more.

Several other simulation frameworks take INET as a base and extend it into specific directions, such as vehicular networks, overlay/peer-to-peer networks, or LTE.

1.2 Scope of this Manual

This manual is written for developers who intend to extend INET with new components written in C++. This manual is accompanied by the INET Reference, which is generated from NED and MSG files using OMNeT++'s documentation generator, and the documentation of the underlying C++ classes, generated from the source files using Doxygen. A working knowledge of OMNeT++ and the C++ language is assumed.

WORKING WITH PACKETS

2.1 Overview

The INET Packet API is designed to facilitate the implementation of communication protocols and applications by providing many useful C++ components. In the following sections, we will introduce the Packet API in detail, and we will shed light on many common API usages through examples.

Note

Code fragments in this chapter have been somewhat simplified for brevity. For example, some `const` modifiers and `const` casts have been omitted, setting fields have been omitted, and some algorithms have been simplified to ease understanding.

The representation of packets is essential for communication network simulation. Applications and communication protocols construct, deconstruct, encapsulate, fragment, aggregate, and manipulate packets in many ways. In order to facilitate the implementation of these behavioral patterns, INET provides a feature-rich general data structure, the `Packet` class.

The `Packet` data structure is capable of representing application packets, TCP segments, IP datagrams, Ethernet frames, IEEE 802.11 frames, and all kinds of digital data. It is designed to provide efficient storage, duplication, sharing, encapsulation, aggregation, fragmentation, serialization, and data representation selection. Additional functionality, such as support for enqueueing data for transmission and buffering received data for reassembly and/or for reordering, is provided as separate C++ data structures on top of `Packet`.

2.2 Representing Data

The `Packet` data structure builds on top of another set of data structures called chunks. Chunks provide several alternatives to represent a piece of data.

INET provides the following built-in chunk C++ classes:

- `Chunk`, the base class for all chunk classes
- repeated byte or bit chunk (`ByteCountChunk`, `BitCountChunk`)
- raw bytes or bits chunk (`BytesChunk`, `BitsChunk`)
- ordered sequence of chunks (`SequenceChunk`)
- slice of another chunk designated by offset and length (`SliceChunk`)
- many protocol specific field based chunks (e.g. `Ipv4Header` subclass of `FieldsChunk`)

In addition, communication protocols and applications often define their own chunk types. User-defined chunks are normally defined in `msg` files as a subclass of `FieldsChunk`, which the OMNeT++ MSG compiler turns into C++ code. It is also possible to write a user defined chunk from scratch.

Chunks usually represent application data and protocol headers. The following examples demonstrate the construction of various chunks.

```

auto bitCountData = makeShared<BitCountChunk>(b(3), 0); // 3 zero bits
auto byteCountData = makeShared<ByteCountChunk>(B(10), '?'); // 10 '?' bytes
auto rawBitsData = makeShared<BitsChunk>();
rawBitsData->setBits({1, 0, 1}); // 3 raw bits
auto rawBytesData = makeShared<BytesChunk>(); // 10 raw bytes
rawBytesData->setBytes({243, 74, 19, 84, 81, 134, 216, 61, 4, 8});
auto fieldBasedHeader = makeShared<UdpHeader>(); // create new UDP header
fieldBasedHeader->setSrcPort(1000); // set some fields

```

In general, chunks must be constructed with a call to the `makeShared` function instead of the standard C++ `new` operator, because chunks are shared among packets using C++ shared pointers.

Packets most often contain several chunks, inserted by different protocols, as they are passed through the protocol layers. The most common way to represent packet contents is to form a compound chunk by concatenation.

```

auto sequence = makeShared<SequenceChunk>(); // create empty sequence
sequence->insertAtBack(makeShared<UdpHeader>()); // append UDP header
sequence->insertAtBack(makeShared<ByteCountChunk>(B(10), 0)); // 10 bytes

```

Protocols often need to slice data, for example, to provide fragmentation, which is also directly supported by the chunk API.

```

auto udpHeader = makeShared<UdpHeader>(); // create 8 bytes UDP header
auto firstHalf = udpHeader->peek(B(0), B(4)); // first 4 bytes of header
auto secondHalf = udpHeader->peek(B(4), B(4)); // second 4 bytes of header

```

In order to avoid cluttered data representation due to slicing, the chunk API provides automatic merging for consecutive chunk slices.

```

auto sequence = makeShared<SequenceChunk>(); // create empty sequence
sequence->insertAtBack(firstHalf); // append first half
sequence->insertAtBack(secondHalf); // append second half
auto merged = sequence->peek(B(0), B(8)); // automatically merge slices

```

Alternative representations can be easily converted into one another using automatic serialization as a common ground.

```

auto raw = merged->peek<BytesChunk>(B(0), B(8)); // auto serialization
auto original = raw->peek<UdpHeader>(B(0), B(8)); // auto deserialization

```

The following MSG fragment is a more complete example which shows how a UDP header could be defined:

```

enum ChecksumMode
{
    CHECKSUM_DISABLED = 0; // CHECKSUM is not set, serializable
    CHECKSUM_DECLARED = 1; // CHECKSUM is correct without the value, not serializable
    CHECKSUM_COMPUTED = 2; // CHECKSUM is potentially incorrect, serializable
}

class UdpHeader extends FieldsChunk
{
    chunkLength = B(8); // UDP header length is always 8 bytes
    int sourcePort = -1; // source port field is undefined by default
    int destinationPort = -1; // destination port field is undefined by default
    B lengthField = B(-1); // length field is undefined by default
    uint16_t checksum = 0; // checksum field is 0 by default
    ChecksumMode checksumMode = CHECKSUM_DISABLED; // checksum mode is disabled by default
}

```

It's important to distinguish the two length related fields in the `UdpHeader` chunk. One is the length of the chunk itself (`chunkLength`), the other is the value in the length field of the header (`lengthField`).

2.3 Representing Packets

The `Packet` data structure uses a single chunk data structure to represent its contents. The contents may be as simple as raw bytes (`BytesChunk`), but most likely, it will be the concatenation (`SequenceChunk`) of various protocol specific headers (e.g., `FieldsChunk` subclasses) and application data (e.g., `ByteCountChunk`).

Packets can be created by both applications and communication protocols. As packets are passed down through the protocol layers at the sender node, new protocol specific headers and trailers are inserted during processing.

```
auto emptyPacket = new Packet("ACK"); // create empty packet
auto data = makeShared<ByteCountChunk>(B(1000));
auto dataPacket = new Packet("DATA", data); // create new packet with data
auto moreData = makeShared<ByteCountChunk>(B(1000));
dataPacket->insertAtBack(moreData); // insert more data at the end
auto udpHeader = makeShared<UdpHeader>(); // create new UDP header
dataPacket->insertAtFront(udpHeader); // insert header into packet
```

In order to facilitate packet processing by communication protocols at the receiver node, `Packet` maintains two offsets into the packet data that divide the data into three regions: front popped part, data part, and back popped part. During packet processing, as the packet is passed through the protocol layers, headers and trailers are popped from the beginning and from the end of the packet, moving the corresponding offsets. This effectively reduces the remaining unprocessed part called the data part, but it doesn't affect the data stored in the packet.

```
packet->popAtFront<MacHeader>(); // pop specific header from packet
packet->popAtBack<MacTrailer>(); // pop specific trailer from packet
auto data = packet->peekData(); // peek remaining data in packet
```

2.4 Representing Signals

Protocols and applications use the `Packet` data structure to represent digital data during the processing within the network node. In contrast, the wireless transmission medium uses a different data structure called `Signal` to represent the physical phenomena used to transmit packets.

```
auto signal = new Signal(transmission);
signal->setDuration(duration);
signal->encapsulate(packet);
```

Signals always encapsulate a packet and also contain a description of the analog domain representation. The most important physical properties of a signal are the signal duration and the signal power.

2.5 Representing Transmission Errors

An essential part of communication network simulation is the understanding of protocol behavior in the presence of errors. The `Packet` API provides several alternatives for representing errors. The alternatives range from simple, but computationally cheap, to accurate, but computationally expensive solutions.

- mark erroneous packets (simple)
- mark erroneous chunks (good compromise)
- change bits in raw chunks (accurate)

The first example shows how to represent transmission errors on the packet level. A packet is marked as erroneous based on its length and the associated bit error rate. This representation doesn't give too much chance for a protocol to do anything else than discard an erroneous packet.

```

Packet *ErrorModel::corruptPacket(Packet *packet, double ber)
{
    auto length = packet->getDataLength();
    auto hasErrors = hasProbabilisticError(length, ber); // decide randomly
    auto corruptedPacket = packet->dup(); // cheap operation
    corruptedPacket->setBitError(hasErrors); // set bit error flag
    return corruptedPacket;
}

```

The second example shows how to represent transmission errors on the chunk level. Similarly to the previous example, a chunk is also marked as erroneous based on its length and the associated bit error rate. This representation allows a protocol to discard only certain parts of the packet. For example, an aggregated packet may be partially discarded and processed.

```

Packet *ErrorModel::corruptChunks(Packet *packet, double ber)
{
    b offset = b(0); // start from the beginning
    auto corruptedPacket = new Packet("Corrupt"); // create new packet
    while (auto chunk = packet->peekAt(offset)->dupShared()) { // for each chunk
        auto length = chunk->getChunkLength();
        auto hasErrors = hasProbabilisticError(length, ber); // decide randomly
        if (hasErrors) // if erroneous
            chunk->markIncorrect(); // set incorrect bit
            corruptedPacket->insertAtBack(chunk); // append chunk to corrupt packet
            offset += chunk->getChunkLength(); // increment offset with chunk length
    }
    return corruptedPacket;
}

```

The last example shows how to actually represent transmission errors on the byte level. In contrast with the previous examples, this time the actual data of the packet is modified. This allows a protocol to discard or correct any part based on checksums.

```

Packet *ErrorModel::corruptBytes(Packet *packet, double ber)
{
    vector<uint8_t> corruptedBytes; // bytes of corrupted packet
    auto data = packet->peekAllAsBytes(); // data of original packet
    for (auto byte : data->getBytes()) { // for each original byte do
        if (hasProbabilisticError(B(1), ber)) // if erroneous
            byte = ~byte; // invert byte (simplified corruption)
            corruptedBytes.push_back(byte); // store byte in corrupted data
    }
    auto corruptedData = makeShared<BytesChunk>(); // create new data
    corruptedData->setBytes(corruptedBytes); // store corrupted bits
    return new Packet("Corrupt", corruptedData); // create new packet
}

```

The physical layer models support the above mentioned different error representations via configurable parameters. Higher layer protocols detect errors by checking the error bit on packets and chunks, and by standard checksum mechanisms.

2.6 Packet Tagging

Within network nodes, supplementary data often needs to be transmitted alongside a packet. For instance, when an application-layer module intends to transfer data using TCP, it must specify a connection identifier for TCP. Similarly, when TCP transmits a segment via IP, IP requires a destination address, and when IP sends a datagram to an Ethernet interface for transmission, a destination MAC address must be specified. These additional details

are attached to a packet as tags.

The following code fragment demonstrates how packet tags could be set in the IPv4 protocol module:

```
void Ipv4::sendDown(Packet *packet, Ipv4Address nextHopAddr, int interfaceId)
{
    auto macAddressReq = packet->addTag<MacAddressReq>(); // add new tag for MAC
    macAddressReq->setSrcAddress(selfAddress); // source is our MAC address
    auto nextHopMacAddress = resolveMacAddress(nextHopAddr); // simplified ARP
    macAddressReq->setDestAddress(nextHopMacAddress); // destination is next hop
    auto interfaceReq = packet->addTag<InterfaceReq>(); // add tag for dispatch
    interfaceReq->setInterfaceId(interfaceId); // set designated interface
    auto packetProtocolTag = packet->addTagIfAbsent<PacketProtocolTag>();
    packetProtocolTag->setProtocol(&Protocol::ipv4); // set protocol of packet
    send(packet, "out"); // send to MAC protocol module of designated interface
}
```

Packet tags are not transmitted from one network node to another. All physical layer protocols delete all packet tags from a packet before sending it to the connected peer or to the transmission medium.

For more details on what kinds of tags are there, see the

2.7 Region Tagging

To gather certain statistics, it might be necessary to add metadata to various regions of packet data. For instance, determining the end-to-end delay of a TCP stream necessitates labeling data regions at the source with their creation timestamp. Subsequently, as the data arrives, the receiver calculates the end-to-end delay for each region.

```
void ClientApp::send()
{
    auto data = makeShared<ByteCountChunk>(); // create new data chunk
    auto creationTimeTag = data->addTag<CreationTimeTag>(); // add new tag
    creationTimeTag->setCreationTime(simTime()); // store current time
    auto packet = new Packet("Data", data); // create new packet
    socket.send(packet); // send packet using TCP socket
}
```

Within a TCP stream, the data may be split, rearranged, and combined in various ways by the underlying network. The packet data representation is responsible for preserving the associated region tags as if they were individually attached to each bit. To prevent a cluttered data representation resulting from the aforementioned characteristics, the tag API offers automatic merging for successive, equivalent tag regions.

```
void ServerApp::receive(Packet *packet)
{
    auto data = packet->peekData(); // get all data from the packet
    auto regions = data->getAllTags<CreationTimeTag>(); // get all tag regions
    for (auto& region : regions) { // for each region do
        auto creationTime = region.getTag()->getCreationTime(); // original time
        auto delay = simTime() - creationTime; // compute delay
        cout << region.getOffset() << region.getLength() << delay; // use data
    }
}
```

The above loop could execute once for the entirety of the data, or it could execute multiple times, depending on the data's creation at the sender and the operation of the underlying network.

2.8 Dissecting Packets

Understanding what's inside a packet is a very important and often used functionality. Simply using the representation may be insufficient, because the `Packet` may be represented with a `BytesChunk`, for example. The `Packet` API provides a `PacketDissector` class which analyzes a packet solely based on the assigned packet protocol and the actual data it contains.

The analysis is done according to the protocol logic as opposed to the actual representation of the data. The `PacketDissector` works similarly to a parser. Basically, it walks through each part (such as protocol headers) of a packet in order. For each part, it determines the corresponding protocol and the most specific representation for that protocol.

The `PacketDissector` class relies on small registered protocol-specific dissector classes (e.g. `Ipv4ProtocolDissector`) subclassing the required `ProtocolDissector` base class. Implementors are expected to use the `PacketDissector::ICallback` interface to notify the parser about the packet structure.

```
void startProtocolDataUnit(Protocol *protocol);
void endProtocolDataUnit(Protocol *protocol);
void markIncorrect();
void visitChunk(Ptr<Chunk>& chunk, Protocol *protocol);
void dissectPacket(Packet *packet, Protocol *protocol);
```

In order to use the `PacketDissector`, the user is expected to implement a `PacketDissector::ICallback` interface. The callback interface will be notified for each part of the packet as the `PacketDissector` goes through it.

```
auto& registry = ProtocolDissectorRegistry::getInstance();
PacketDissector dissector(registry, callback);
auto packetProtocolTag = packet->findTag<PacketProtocolTag>();
auto protocol = packetProtocolTag->getProtocol();
dissector.dissectPacket(packet, protocol);
```

2.9 Filtering Packets

Filtering packets based on the actual data they contain is another widely used and very important feature. With the help of the packet dissector, it is straightforward to create arbitrary custom packet filters. Packet filters are generally used for recording packets and visualizing various packet-related information.

In order to simplify filtering, the `Packet` API provides a generic expression-based packet filter that is implemented in the `PacketFilter` class. The expression syntax is the same as other OMNeT++ expressions, and the data filter is matched against individual chunks of the packet as found by the packet dissector.

For example, the packet filter expression “ping*” matches all packets having the name prefix ‘ping’, and the packet chunk filter expression “inet::Ipv4Header and srcAddress(10.0.0.*)” matches all packets that contain an IPv4 header with a ‘10.0.0’ source address prefix.

```
PacketFilter filter; // patterns for the whole packet and for the data
filter.setPattern("ping*", "Ipv4Header and sourceAddress(10.0.0.*)");
filter.matches(packet); // returns boolean value
```

2.10 Printing Packets

During model development, packets often need to be displayed in a human-readable form. The `Packet` API provides a `PacketPrinter` class that is capable of forming a human-readable string representation of `Packet`'s. The `PacketPrinter` class relies on small registered protocol-specific printer classes (e.g. `Ipv4ProtocolPrinter`) subclassing the required `ProtocolPrinter` base class.

The packet printer is automatically used by the OMNeT++ runtime user interface to display packets in the packet log window. The packet printer contributes several log window columns into the user interface: ‘Source’, ‘Destination’,

'Protocol', 'Length', and 'Info'. These columns display packet data similarly to the well-known Wireshark protocol analyzer.

```
PacketPrinter printer; // turns packets into human readable strings
printer.printPacket(std::cout, packet); // print to standard output
```

The PacketPrinter provides a few other functions that have additional options to control the details of the resulting human readable form.

2.11 Recording PCAP

Exporting the packets from a simulation into a PCAP file allows further processing with 3rd party tools. The Packet API provides a PcapDump class for creating PCAP files. Packet filtering can be used to reduce the file size and increase performance.

```
PcapDump dump;
dump.openPcap("out.pcap", 65535, 0); // maximum length and PCAP type
dump.writePacket(simTime(), packet); // record with current time
```

2.12 Encapsulating Packets

Many communication protocols work with simple packet encapsulation. They encapsulate packets with their own protocol-specific headers and trailers at the sender node, and they decapsulate packets at the receiver node. The headers and trailers carry the information that is required to provide the protocol specific service.

For example, the Ethernet MAC protocol encapsulates an IP datagram by prepending the packet with an Ethernet MAC header and also by appending the packet with an optional padding and an Ethernet FCS. The following example shows how a MAC protocol could encapsulate a packet:

```
void Mac::encapsulate(Packet *packet)
{
    auto header = makeShared<MacHeader>(); // create new header
    header->setChunkLength(B(8)); // set chunk length to 8 bytes
    header->setLengthField(packet->getDataLength()); // set length field
    header->setTransmitterAddress(selfAddress); // set other header fields
    packet->insertAtFront(header); // insert header into packet
    auto trailer = makeShared<MacTrailer>(); // create new trailer
    trailer->setChunkLength(B(4)); // set chunk length to 4 bytes
    trailer->setFcsMode(FCS_MODE_DECLARED); // set trailer fields
    packet->insertAtBack(trailer); // insert trailer into packet
}
```

When receiving a packet, the Ethernet MAC protocol removes an Ethernet MAC header and an Ethernet FCS from the packet and passes the resulting IP datagram along. The following example shows how a MAC protocol could decapsulate a packet:

```
void Mac::decapsulate(Packet *packet)
{
    auto header = packet->popAtFront<MacHeader>(); // pop header from packet
    auto lengthField = header->getLengthField();
    cout << header->getChunkLength() << endl; // print chunk length
    cout << lengthField << endl; // print header length field
    cout << header->getReceiverAddress() << endl; // print other header fields
    auto trailer = packet->popAtBack<MacTrailer>(); // pop trailer from packet
    cout << trailer->getFcsMode() << endl; // print trailer fields
    assert(packet->getDataLength() == lengthField); // if the packet is correct
}
```

Although the `popAtFront` and `popAtBack` functions change the remaining unprocessed part of the packet, they do not have an effect on the actual packet data. That is when the packet reaches high-level protocol, it still contains all the received data, but the remaining unprocessed part is smaller.

2.13 Fragmenting Packets

Communication protocols often provide fragmentation to overcome various physical limits (e.g. length limit, error rate). They split packets into smaller pieces at the sender node, which send them one-by-one. They form the original packet at the receiver node by combining the received fragments.

For example, the IEEE 802.11 protocol fragments packets to overcome the increasing probability of packet loss of large packets. The following example shows how a MAC protocol could fragment a packet:

```
vector<Packet *> *Mac::fragment(Packet *packet, vector<b>& sizes)
{
    auto offset = b(0); // start from the packet's beginning
    auto fragments = new vector<Packet *>(); // result collection
    for (auto size : sizes) { // for each received size do
        auto fragment = new Packet("Fragment"); // header + data part + trailer
        auto header = makeShared<MacHeader>(); // create new header
        header->setFragmentOffset(offset); // set fragment offset for reassembly
        fragment->insertAtFront(header); // insert header into fragment
        auto data = packet->peekAt(offset, size); // get data part from packet
        fragment->insertAtBack(data); // insert data part into fragment
        auto trailer = makeShared<MacTrailer>(); // create new trailer
        fragment->insertAtBack(trailer); // insert trailer into fragment
        fragments->push_back(fragment); // collect fragment into result
        offset += size; // increment offset with size of data part
    }
    return fragments;
}
```

When receiving fragments, protocols need to collect the coherent fragments of the same packet until all fragments become available. The following example shows how a MAC protocol could form the original packet from a set of coherent fragments:

```
Packet *Mac::defragment(vector<Packet *>& fragments)
{
    auto packet = new Packet("Original"); // create new concatenated packet
    for (auto fragment : fragments) {
        fragment->popAtFront<MacHeader>(); // pop header from fragment
        fragment->popAtBack<MacTrailer>(); // pop trailer from fragment
        packet->insertAtBack(fragment->peekData()); // concatenate fragment data
    }
    return packet;
}
```

2.14 Aggregating Packets

Communication protocols often provide aggregation to better utilize the communication channel by reducing protocol overhead. They wait for several packets to arrive at the sender node, then they form a large aggregated packet which is in turn sent at once. At the receiver node, the aggregated packet is split into the original packets, and they are passed along.

For example, the IEEE 802.11 protocol aggregates packets for better channel utilization at both MSDU and MPDU levels. The following example shows how a MAC protocol could create an aggregate packet:


```

Packet *Mac::aggregate(vector<Packet *>& packets)
{
    auto aggregate = new Packet("Aggregate"); // create concatenated packet
    for (auto packet : packets) { // for each received packet do
        auto header = makeShared<SubHeader>(); // create new subheader
        header->setLengthField(packet->getDataLength()); // set subframe length
        aggregate->insertAtBack(header); // insert subheader into aggregate
        auto data = packet->peekData(); // get packet data
        aggregate->insertAtBack(data); // insert data into aggregate
    }
    auto header = makeShared<MacHeader>(); // create new header
    header->setAggregate(true); // set aggregate flag
    aggregate->insertAtFront(header); // insert header into aggregate
    auto trailer = makeShared<MacTrailer>(); // create new trailer
    aggregate->insertAtBack(trailer); // insert trailer into aggregate
    return aggregate;
}

```

The following example shows how a MAC protocol could disaggregate a packet:

```

vector<Packet *> *Mac::disaggregate(Packet *aggregate)
{
    aggregate->popAtFront<MacHeader>(); // pop header from packet
    aggregate->popAtBack<MacTrailer>(); // pop trailer from packet
    vector<Packet *> *packets = new vector<Packet *>(); // result collection
    b offset = aggregate->getFrontOffset(); // start after header
    while (offset != aggregate->getBackOffset()) { // up to trailer
        auto header = aggregate->peekAt<SubHeader>(offset); // peek sub header
        offset += header->getChunkLength(); // increment with header length
        auto size = header->getLengthField(); // get length field from header
        auto data = aggregate->peekAt(offset, size); // peek following data part
        auto packet = new Packet("Original"); // create new packet
        packet->insertAtBack(data); // insert data into packet
        packets->push_back(packet); // collect packet into result
        offset += size; // increment offset with data size
    }
    return packets;
}

```

2.15 Serializing Packets

In real communication systems, packets are usually stored as a sequence of bytes directly in network byte order. In contrast, INET usually stores packets in small field-based C++ classes (generated by the OMNeT++ MSG compiler) to ease debugging. In order to calculate checksums or to communicate with real hardware, all protocol-specific parts must be serializable to a sequence of bytes.

The protocol header serializers are separate classes from the actual protocol headers. They must be registered in the ChunkSerializerRegistry in order to be used. The following example shows how a MAC protocol header could be serialized to a sequence of bytes:

```

void MacHeaderSerializer::serialize
(MemoryOutputStream& stream, Ptr<Chunk>& chunk)
{
    auto header = staticPtrCast<MacHeader>(chunk);
    stream.writeUInt16Be(header->getType()); // unsigned 16 bits, big endian
    stream.writeMacAddress(header->getTransmitterAddress());
}

```

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```

stream.writeMacAddress(header->getReceiverAddress());
}

```

Deserialization is somewhat more complicated than serialization because it must be prepared to handle incomplete or even incorrect data due to errors introduced by the network. The following example shows how a MAC protocol header could be deserialized from a sequence of bytes:

```

Ptr<Chunk> MacHeaderSerializer::deserialize(MemoryInputStream& stream)
{
    auto header = makeShared<MacHeader>(); // create new header
    header->setType(stream.readUInt16Be()); // unsigned 16 bits, big endian
    header->setTransmitterAddress(stream.readMacAddress());
    header->setReceiverAddress(stream.readMacAddress());
    return header;
}

```

2.16 Emulation Support

In order to be able to communicate with real hardware, packets must be converted to and from a sequence of bytes. The reason is that the programming interface of operating systems and external libraries work with sending and receiving raw data.

All protocol headers and data chunks that are present in a packet must have a registered serializer to be able to create the raw sequence of bytes. Protocol modules must also be configured to either disable or compute checksums because serializers cannot carry out the checksum calculation.

The following example shows how a packet could be converted to a sequence of bytes to send through an external interface:

```

vector<uint8_t>& ExternalInterface::prepareToSend(Packet *packet)
{
    auto data = packet->peekAllAsBytes(); // convert to a sequence of bytes
    return data->getBytes(); // actual bytes to send
}

```

The following example shows how a packet could be converted from a sequence of bytes when receiving from an external interface:

```

Packet *ExternalInterface::prepareToReceive(vector<uint8_t>& bytes)
{
    auto data = makeShared<BytesChunk>(bytes); // create chunk with bytes
    return new Packet("Emulation", data); // create packet with data
}

```

In INET, all protocols automatically support hardware emulation due to the dual representation of packets. The above example creates a packet that contains a single chunk with a sequence of bytes. As the packet is passed through the protocols, they can interpret the data (e.g., by calling `peekAtFront`) as they see fit. The Packet API always provides the requested representation, either because it's already available in the packet or because it gets automatically deserialized.

2.17 Queueing Packets

Some protocols store packet data temporarily at the sender node before actual processing can occur. For example, the TCP protocol must store the outgoing data received from the application to be able to provide transmission flow control.

The following example shows how a transport protocol could store the received data temporarily until the data is actually used:

```
class TransportSendQueue
{
    ChunkQueue queue; // stores application data
    B sequenceNumber; // position in stream

    void enqueueApplicationData(Packet *packet);
    Packet *createSegment(b length);
};

void TransportSendQueue::enqueueApplicationData(Packet *packet)
{
    queue.push(packet->peekData()); // store received data
}

Packet *TransportSendQueue::createSegment(b maxLength)
{
    auto packet = new Packet("Segment"); // create new segment
    auto header = makeShared<TransportHeader>(); // create new header
    header->setSequenceNumber(sequenceNumber); // store sequence number for reordering
    packet->insertAtFront(header); // insert header into segment
    if (queue.getLength() < maxLength)
        maxLength = queue.getLength(); // reduce length if necessary
    auto data = queue.pop(maxLength); // pop requested amount of data
    packet->insertAtBack(data); // insert data into segment
    sequenceNumber += data->getChunkLength(); // increase sequence number
    return packet;
}
```

The `ChunkQueue` class acts similarly to a binary FIFO queue except it works with chunks. Similarly to the `Packet`, it also automatically merges consecutive data and selects the most appropriate representation.

2.18 Buffering Packets

Protocols at the receiver node often need to buffer incoming packet data until the actual processing can occur. For example, packets may arrive out of order, and the data they contain must be reassembled or reordered before it can be passed along.

INET provides a few special-purpose C++ classes to support data buffering:

- `ChunkBuffer` provides automatic merging for large data chunks from out of order smaller data chunks.
- `ReassemblyBuffer` provides reassembling for out-of-order data according to an expected length.
- `ReorderBuffer` provides reordering for out-of-order data into a continuous data stream from an expected offset.

All buffers deal with only the data, represented by chunks, instead of packets. They automatically merge consecutive data and select the most appropriate representation. Protocols using these buffers automatically support all data representations provided by INET, and any combination thereof. For example, `ByteCountChunk`, `BytesChunk`, `FieldsChunk`, and `SliceChunk` can be freely mixed in the same buffer.

2.19 Reassembling Packets

Some protocols may use an unreliable service to transfer a large piece of data over the network. The unreliable service requires the receiver node to be prepared for receiving parts out of order and potentially duplicated.

For example, the IP protocol must store incoming fragments at the receiver node because it must wait until the datagram becomes complete before it can be passed along. The IP protocol must also be prepared for receiving the individual fragments out of order and potentially duplicated.

The following example shows how a network protocol could store and reassemble the data of the incoming packets into a whole packet:

```
class NetworkProtocolDefragmentation
{
    ReassemblyBuffer buffer; // stores received data

    void processDatagram(Packet *packet); // processes incoming packets
    Packet *getReassembledDatagram(); // reassembles the original packet
};

void NetworkProtocolDefragmentation::processDatagram(Packet *packet)
{
    auto header = packet->popAtFront<NetworkProtocolHeader>(); // remove header
    auto fragmentOffset = header->getFragmentOffset(); // determine offset
    auto data = packet->peekData(); // get data from packet
    buffer.replace(fragmentOffset, data); // overwrite data in buffer
}

Packet *NetworkProtocolDefragmentation::getReassembledDatagram()
{
    if (!buffer.isComplete()) // if reassembly isn't complete
        return nullptr; // there's nothing to return
    auto data = buffer.getReassembledData(); // complete reassembly
    return new Packet("Datagram", data); // create new packet
}
```

The ReassemblyBuffer supports replacing the stored data at a given offset, and it also provides the complete reassembled data with the expected length if available.

2.20 Reordering Packets

Some protocols may use an unreliable service to transfer a long data stream over the network. The unreliable service requires the sender node to resend unacknowledged parts, and it also requires the receiver node to be prepared for receiving parts out of order and potentially duplicated.

For example, the TCP protocol must buffer the incoming data at the receiver node because the TCP segments may arrive out of order and potentially duplicated or overlapping, and TCP is required to provide the data to the application in the correct order and only once.

The following example shows how a transport protocol could store and reorder the data of incoming packets, which may arrive out of order, and also how such a protocol could pass along only the available data in the correct order:

```
class TransportReceiveQueue
{
    ReorderBuffer buffer; // stores receive data
    B sequenceNumber;

    void processSegment(Packet *packet);
    Packet *getAvailableData();
};

void TransportReceiveQueue::processSegment(Packet *packet)
{
    auto header = packet->popAtFront<TransportHeader>(); // pop transport header
```

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```
    auto sequenceNumber = header->getSequenceNumber();
    auto data = packet->peekData(); // get all packet data
    buffer.replace(sequenceNumber, data); // overwrite data in buffer
}

Packet *TransportReceiveQueue::getAvailableData()
{
    if (buffer.getAvailableDataLength() == 0) // if no data available
        return nullptr;
    auto data = buffer.popAvailableData(); // remove all available data
    return new Packet("Data", data);
}
```

The ReorderBuffer supports replacing the stored data at a given offset, and it provides the available data from the expected offset if any.

COMMUNICATING WITH TAGS

3.1 Overview

Modules often exchange information by sending packets along with supplementary data, referred to as tags. A tag is usually a small data structure that focuses on a single aspect of a protocol's parameterization. Tags can be attached to the whole packet, known as packet tags, or to specific parts of the packet, known as region tags. Tags are implemented as C++ classes that serve as data containers and are usually generated by the OMNeT++ MSG compiler. The primary types of tags are as follows:

- *Requests* carry information from higher layers to lower layers (e.g., `MacAddressReq`).
- *Indications* carry information from lower layers to higher layers (e.g., `InterfaceInd`).
- *Plain tags* contain meta-information (e.g., `PacketProtocolTag`).
- *Base classes* must not be attached to packets (e.g., `TagBase`).

For example, a request tag that specifies the source and destination MAC address could be implemented in an MSG file as shown below:

```
class MacAddressReq extends TagBase
{
    MacAddress srcAddress; // may be unspecified
    MacAddress destAddress; // always specified
}
```

The following is a short description of several commonly used packet tags:

- `PacketProtocolTag`: Specifies the protocol of the packet's contents.
- `DispatchProtocolReq`: Specifies the receiver protocol module inside the network node.
- `EncapsulationProtocolReq`: Specifies the requested protocol header encapsulation order.
- `SocketReq`: Specifies the application socket.
- `L4PortReq`: Specifies the source and destination ports.
- `L3AddressReq`: Specifies source and destination network addresses.
- `InterfaceReq`: Specifies the outgoing network interface.
- `NextHopAddressReq`: Specifies the next-hop address for packet routing.
- `VlanReq`: Specifies the virtual LAN identifier of IEEE 802.1Q.
- `PcpReq`: Specifies the priority code point of IEEE 802.1Q.
- `StreamReq`: Specifies the TSN stream identifier inside the network node.
- `MacAddressReq`: Specifies source and destination MAC addresses.
- `Ieee80211ModeReq`: Specifies the IEEE 802.11 PHY mode.
- `Ieee80211ChannelReq`: Specifies the IEEE 802.11 channel.

- `SignalPowerReq`: Specifies transmit signal power.

All request tags have their corresponding indication tags. For example, there are indications such as `SocketInd`, `InterfaceInd`, and `StreamInd`. The requests are usually attached to outgoing packets, while the indications are usually attached to incoming packets.

The following is a short description of several commonly used region tags:

- `IdentityTag`: Uniquely identifies individual bits in the network over the lifetime of the whole simulation.
- `CreationTimeTag`: Specifies the creation time of data regions for lifetime measurements.
- `FlowTag`: Specifies the packet flows of data regions for various flow-specific measurements.
- `PacketEventTag`: Carries information about queueing, processing, transmission, etc. events that happened to data regions.

3.2 Communicating Through Protocol Layers

Tags can pass through protocol modules, potentially reaching far beyond the module that initially attached them, in both the downward and upward directions. Typically, tags are removed at the point where they are processed, either by being transformed into header fields within a packet or used for some protocol-specific decision-making. Protocols have the freedom to disregard any tags based on their configuration and state.

Both packet tags and region tags usually remain unchanged for many operations that protocol modules carry out with packets. For example, when packets are enqueued/dequeued, encapsulated/decapsulated, cloned, buffered, or stored for later reuse, the tags remain unchanged.

3.3 Specifying the Protocol of a Packet

The most important packet tag is the `PacketProtocolTag`. It specifies the outermost protocol of the packet. This tag should always be present because the packet protocol cannot be correctly determined just by looking at the raw data. In contrast, the inner protocol headers in the packet can usually be recursively identified by protocol fields such as the protocol ID field of the IPv4 header. The `PacketProtocolTag` is used, among other things: - for dissecting the packet along the protocol headers, - for printing the packet as a human-readable string to help interpret its contents.

Normally, a packet is transformed from one protocol to another in a single step, so the packet protocol tag either specifies the protocol before the operation or the protocol after the operation. For example, the `Udp` protocol module encapsulates the outgoing packet using a `UdpHeader`. The packet protocol is set to an application-specific protocol before the UDP encapsulation, and it's set to the UDP protocol after the encapsulation.

Sometimes, protocol implementations themselves are split up into several smaller modules. For example, the modular Ethernet implementation uses a separate module for the insertion of the Ethernet MAC header and the Ethernet FCS. In such a case, the packet protocol tag can only specify the inner protocol that is being encapsulated into an Ethernet MAC frame.

3.4 Dispatching Packets to Protocol Modules

Inside a network node, protocol modules interact with one another by sending `Packet` or `Message` objects. INET provides great flexibility in terms of how the protocol modules can be connected. Protocols can be connected directly to each other or they can be connected through one or more `MessageDispatcher` modules. This flexibility allows for the creation of simple as well as complex network node architectures.

3.4.1 How to Connect Protocol Modules

Simple network nodes can be constructed using a linear protocol stack, where protocol modules are directly connected to one another without using message dispatcher modules.

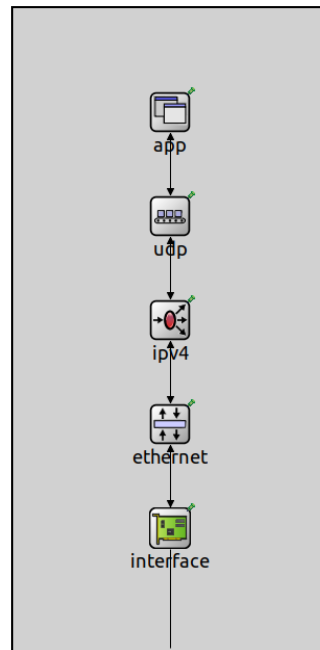


Fig. 1: Simple network node structure in the IDE

More complex network nodes can be created by grouping protocols into layers and connecting them through `MessageDispatcher` modules, which facilitate many-to-one and many-to-many relationships among the protocols of the layers.

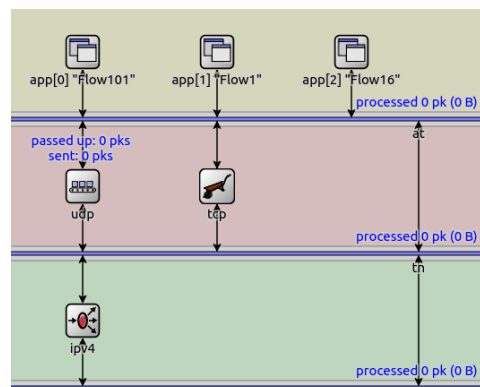


Fig. 2: Complex network node structure in QtEnv

It's also possible to use message dispatcher modules hierarchically within multiple levels of nested compound modules. Ultimately, one could even connect all protocols to a single central message dispatcher module. However, please note that only one instance of a given protocol module can be connected to a message dispatcher.

To support the packet dispatching mechanism, certain additional requirements must be met in the C++ code:

- Protocols must be registered using the `registerProtocol` function.
- Packets must have the `DispatchProtocolReq` tags attached.

3.4.2 Registering Protocols

Protocol modules must call the `registerProtocol` function from the `initialize` method to inform connected `MessageDispatcher` modules of their presence. The following code fragment demonstrates this for the IPv4 protocol implementation:

```
void Ipv4::initialize(int stage)
{
    if (stage == INITSTAGE_NETWORK_LAYER) {
        registerService(Protocol::ipv4, gate("transportIn"), gate("transportOut"));
        registerProtocol(Protocol::ipv4, gate("queueOut"), gate("queueIn"));
    }
}
```

Registering the protocols allows the dispatcher modules to learn which gates the protocol modules are connected to. The same protocol is not allowed to be registered in the same message dispatcher using different gates because that would make the dispatching mechanism ambiguous.

3.4.3 Sending Packets with Dispatch Request

In order for the message dispatcher modules to correctly dispatch packets to the intended recipient within the network node, packets and messages must have the `DispatchProtocolReq` tag attached. The following example shows how a MAC protocol could send up a packet to the `Ipv4` protocol module without actually knowing where that module is connected in the network node architecture:

```
void Mac::sendUp(Packet *packet)
{
    auto req = packet->addTagIfAbsent<DispatchProtocolReq>();
    req->setProtocol(&Protocol::ipv4); // set destination protocol
    req->setServicePrimitive(SP_INDICATION); // determine receiving gate
    send(packet, "upperLayerOut");
}
```

The `DispatchProtocolReq` tag specifies both the intended recipient protocol and the requested service primitive. The service primitive, similarly to OSI terminology, can be one of the following:

- `SP_REQUEST` for service requests from layer N+1 to layer N
- `SP_CONFIRM` for service confirmations from layer N to layer N+1
- `SP_INDICATION` for protocol indications from layer N to layer N+1
- `SP_RESPONSE` for protocol response from layer N+1 to layer N

Currently, INET modules only use the `SP_REQUEST` and `SP_INDICATION` service primitives; the other two are only present for completeness. The request service primitive is used when a higher layer protocol module (e.g., `Tcp`) wants to deliver a packet to a lower layer protocol module (e.g., `Ipv4`). Similarly, the `SP_INDICATION` service primitive is used when a lower layer protocol module (e.g., Ethernet) wants to deliver a packet to a higher layer protocol module (e.g., `Ipv4`).

3.5 Determining the Next Protocol

A protocol module has several options for determining which protocol to forward a packet to. Some possibilities include:

- Hard-coding the next protocol in C++. For example, the UDP protocol is hard-coded in C++ in the `UdpSocket` class, and similarly, other protocols are also hard-coded in other protocol-specific sockets.
- Specifying the next protocol using a module parameter. For instance, a network interface module can specify its expected protocol. The `Ipv4` module uses this information to dispatch a packet to the expected protocol of the selected route's network interface.
- Determining the next protocol based on module state. For example, the TSN stream encoder module forwards packets that match the TSN stream mapping to the 802.1 Q-TAG protocol for encapsulation.
- Determining the next protocol based on a packet header field. The `Ipv4` module, for example, uses the IP protocol ID header field from the `Ipv4Header` to look up the next protocol, as shown below:

```
const Protocol *Ipv4::getNextProtocol(Packet *packet)
{
    auto ipv4Header = packet->peekAtFront<Ipv4Header>();
    auto ipProtocolId = ipv4Header->getProtocolId();
    return ProtocolGroup::getIpProtocolGroup()->getProtocol(ipProtocolId);
}
```

- Determining the next protocol based on packet metadata. For instance, the `Tcp` module uses the type of the destination address from the `L3AddressReq` tag to determine whether the packet should be sent to the `Ipv4` or `Ipv6` module.
- Indirectly specifying the next protocol via a protocol encapsulation request. Other modules may have attached an `EncapsulationProtocolReq` to the packet in an earlier stage of the packet processing.

3.6 Controlling the Packet Encapsulation Order

A packet typically contains multiple protocol-specific headers, such as TCP, IP, Ethernet, and sometimes additional optional headers like 802.1Q, 802.1R, 802.1AE, and others. The order of the packet headers is determined by the order in which the packet reaches the relevant protocol modules for encapsulation.

The encapsulation process may need to be different for each packet. For example, an application may need to send a packet to a specific VLAN. In this case, the application should attach a `VlanReq` tag to the packet with the desired VLAN ID. However, it cannot directly send the packet to the relevant 802.1Q protocol module because the packet may need to be delivered to the UDP protocol first. To achieve the desired protocol encapsulation order, the application should also attach an `EncapsulationProtocolReq` tag that specifies that the packet should ultimately be delivered to the 802.1Q protocol for encapsulation. The underlying protocol modules will use this information to determine when the 802.1Q encapsulation should take place.

The `EncapsulationProtocolReq` generally outlines the sequence of protocol modules that a packet should be delivered to for further encapsulation. Additional tags attached to the packet are used as additional parameters for the requested processing steps. The attached encapsulation request may be changed several times during packet processing; new protocols may be added, already added protocols may be removed, etc.

For example, the IP protocol determines the outgoing interface using the routing table and the destination address. The selected network interface specifies the expected protocol that the packet should have in order for the interface module to operate properly. The specified protocol is appended to the end of the requested encapsulation protocols of the packet because it should be the last encapsulation before the packet reaches the network interface. For example, if the IP module selects an Ethernet network interface, then it appends the Ethernet MAC protocol to the `EncapsulationProtocolReq`. As a result, the packet is ultimately encapsulated into an `EthernetMacHeader` before reaching the network interface module.

3.7 Transforming Inbound Packets to Outbound

As part of the forwarding process of Ethernet switches, an inbound packet is transformed into an outbound packet. By default, this process is carried out by the `PacketDirectionReverser` module. The transformation is more like a policy and can be replaced by the user with other modules. The default module doesn't change the packet contents, except for removing the already popped front and back parts. However, it changes the attached packet tags significantly.

The inbound packet usually contains a few tags such as `PacketProtocolTag` and `DirectionTag`, and it also contains several indications such as the `InterfaceInd`, `MacAddressInd`, `VlanInd`, `PcpInd`, `EncapsulationProtocolInd`, etc. During the transformation, only the `PacketProtocolTag` is kept, and all attached indications are removed. Additionally, a set of requests is attached to the packet so that the packet will be encapsulated in the same protocol headers and sent out on the same interface as it came in.

Of course, this is just the start of the processing of the outbound packet. During the several steps that follow, any of the attached requests can be replaced with new ones, potentially resulting in the packet being handled in a completely different way.

USING SOCKETS

4.1 Overview

The INET Socket API provides special C++ abstractions on top of the standard OMNeT++ message passing interface for several communication protocols.

Sockets are most often used by applications and routing protocols to access the corresponding protocol services. Sockets are capable of communicating with the underlying protocol in a bidirectional way. They can assemble and send service requests and packets, and they can also receive service indications and packets.

Applications can simply call the socket class member functions (e.g. `bind()`, `connect()`, `send()`, `close()`) to create and configure sockets, and to send and receive packets. They may also use several different sockets simultaneously.

The following sections first introduce the shared functionality of sockets, and then list all INET sockets in detail, mostly by shedding light on many common usages through examples.

Note

Code fragments in this chapter have been somewhat simplified for brevity. For example, some **virtual** modifiers and **override** qualifiers have been omitted, and some algorithms have been simplified to ease understanding.

4.1.1 Socket Interfaces

Although sockets are always implemented as protocol specific C++ classes, INET also provides C++ socket interfaces. These interfaces allow writing general C++ code which can handle many different kinds of sockets all at once.

For example, the `ISocket` interface is implemented by all sockets, and the `INetworkSocket` interface is implemented by all network protocol sockets.

4.1.2 Identifying Sockets

All sockets have a socket identifier which is unique within the network node. It is automatically assigned to the sockets when they are created. The identifier can be accessed with `getSocketId()` throughout the lifetime of the socket.

The socket identifier is also passed along in `SocketReq` and `SocketInd` packet tags. These tags allow applications and protocols to identify the socket to which `Packet`'s, service `Request`'s, and service `Indication`'s belong.

4.1.3 Configuring Sockets

Since all sockets work with message passing under the hoods, they must be configured prior to use. In order to send packets and service requests on the correct gate towards the underlying communication protocol, the output gate must be configured:

```
socket.setOutputGate(gate("socketOut")); // configure socket output gate
socket.setCallback(this); // set callback interface for message processing
```

In contrast, incoming messages such as service indications from the underlying communication protocol can be received on any application gate.

To ease application development, all sockets support storing a user specified data object pointer. The pointer is accessible with the `setUserData()`, `getUserData()` member functions.

Another mandatory configuration for all sockets is setting the socket callback interface. The callback interface is covered in more detail in the following section.

Other socket specific configuration options are also available, these are discussed in the section of the corresponding socket.

4.1.4 Callback Interfaces

To ease centralized message processing, all sockets provide a callback interface which must be implemented by applications. The callback interface is usually called `ICallback`, and it's defined as an inner class of the socket it belongs to. These interfaces often contain some generic notification methods along with several socket specific methods.

For example, the most common callback method is the one which processes incoming packets:

```
class ICallback // usually the inner class of the socket
{
    void socketDataArrived(ISocket *socket, Packet *packet);
};
```

4.1.5 Processing Messages

In general, sockets can process all incoming messages which were sent by the underlying protocol. The received messages must be processed by the socket where they belong to.

For example, an application can simply go through each known socket in any order, and decide which one should process the received message as follows:

```
if (socket.belongsToSocket(message)) // match message and socket
    socket.processMessage(message); // invoke callback interface
```

Sockets usually deconstruct the received messages and update their state accordingly if necessary. They also automatically dispatch received packets and service indications for further processing to the appropriate functions in the corresponding `ICallback` interface.

4.1.6 Sending Data

All sockets provide one or more `send()` functions which send packets using the current configuration of the socket. The actual means of packet delivery depends on the underlying communication protocol, but in general the state of the socket is expected to affect it.

For example, after the socket is properly configured, the application can start sending packets without attaching any tags, because the socket takes care of the necessary technical details:

```
socket.send(packet); // by means of the underlying communication protocol
```

4.1.7 Receiving Data

For example, the application may directly implement the `ICallback` interface of the socket and print the received data as follows:

```

class App : public cSimpleModule, public ICallback
{
    void socketDataArrived(ISocket *socket, Packet *packet);
};

void App::socketDataArrived(ISocket *socket, Packet *packet)
{
    EV << packet->peekData() << endl;
}

```

4.1.8 Closing Sockets

Sockets must be closed before deleting them. Closing a socket allows the underlying communication protocol to release allocated resources. These resources are often allocated on the local network node, the remote network node, or potentially somewhere else in the network.

For example, a socket for a connection oriented protocol must be closed to release the allocated resources at the peer:

```
socket.close(); // release allocated local and remote network resources
```

4.1.9 Using Multiple Sockets

If the application needs to manage a large number of sockets, for example in a server application which handles multiple incoming connections, the generic `SocketMap` class may be useful. This class can manage all kinds of sockets which implement the `ISocket` interface simultaneously.

For example, processing an incoming packet or service indication can be done as follows:

```

auto socket = socketMap.findSocketFor(message); // lookup socket to process
socket->processMessage(message); // dispatch message to callback interface

```

In order for the `SocketMap` to operate properly, sockets must be added to and removed from it using the `addSocket()` and `removeSocket()` methods respectively.

4.2 UDP Socket

The `UdpSocket` class provides an easy to use C++ interface to send and receive UDP datagrams. The underlying UDP protocol is implemented in the `Udp` module.

4.2.1 Callback Interface

Processing packets and indications which are received from the `Udp` module is pretty simple. The incoming message must be processed by the socket where it belongs as shown in the general section.

The `UdpSocket` deconstructs the message and uses the `UdpSocket::ICallback` interface to notify the application about received data and error indications:

```

class ICallback // inner class of UdpSocket
{
    void socketDataArrived(UdpSocket *socket, Packet *packet);
    void socketErrorArrived(UdpSocket *socket, Indication *indication);
};

```

4.2.2 Configuring Sockets

For receiving UDP datagrams on a socket, it must be bound to an address and a port. Both the address and port are optional. If the address is unspecified, then all UDP datagrams with any destination address are received. If the port is -1, then an unused port is selected automatically by the `Udp` module. The address and port pair must be unique within the same network node.

Here is how to bind to a specific local address and port to receive UDP datagrams:

```
socket.bind(Ipv4Address("10.0.0.42"), 42); // local address/port
```

For only receiving UDP datagrams from a specific remote address/port, the socket can be connected to the desired remote address/port:

```
socket.connect(Ipv4Address("10.0.0.42"), 42); // remote address/port
```

There are several other socket options (e.g. receiving broadcasts, managing multicast groups, setting type of service) which can also be configured using the `UdpSocket` class:

```
socket.setTimeToLive(16); // change default TTL
socket.setBroadcast(true); // receive all broadcasts
socket.joinMulticastGroup(Ipv4Address("224.0.0.9")); // receive multicasts
```

4.2.3 Sending Data

After the socket has been configured, applications can send datagrams to a remote address and port via a simple function call:

```
socket.sendTo(packet42, Ipv4Address("10.0.0.42"), 42); // remote address/port
```

If the application wants to send several datagrams, it can optionally connect to the destination.

The UDP protocol is in fact connectionless, so when the `Udp` module receives the connect request, it simply remembers the remote address and port, and uses it as the default destination for later sends.

```
socket.connect(Ipv4Address("10.0.0.42"), 42); // remote address/port
socket.send(packet1); // send packets via connected socket
// ...
socket.send(packet42);
```

The application can call connect several times on the same socket.

4.2.4 Receiving Data

For example, the application may directly implement the `UdpSocket::ICallback` interface and print the received data as follows:

```
class UdpApp : public cSimpleModule, public UdpSocket::ICallback
{
    void socketDataArrived(UdpSocket *socket, Packet *packet);
};

void UdpApp::socketDataArrived(UdpSocket *socket, Packet *packet)
{
    EV << packet->peekData() << endl;
}
```


4.3 TCP Socket

The `TcpSocket` class provides an easy to use C++ interface to manage TCP connections, and to send and receive data. The underlying TCP protocol is implemented in the `Tcp`, `TcpLwip`, and `TcpNsc` modules.

4.3.1 Callback Interface

Messages received from the various `Tcp` modules can be processed by the `TcpSocket` where they belong to. The `TcpSocket` deconstructs the message and uses the `TcpSocket::ICallback` interface to notify the application about the received data or service indication:

```
class ICallback // inner class of TcpSocket
{
    void socketDataArrived(TcpSocket* socket, Packet *packet, bool urgent);
    void socketAvailable(TcpSocket *socket, TcpAvailableInfo *info);
    void socketEstablished(TcpSocket *socket);
    // ...
    void socketClosed(TcpSocket *socket);
    void socketFailure(TcpSocket *socket, int code);
};
```

4.3.2 Configuring Connections

The `Tcp` module supports several different TCP congestion algorithms, which can also be configured using the `TcpSocket`:

```
socket.setTCPAlgorithmClass("TcpReno");
```

Upon setting the individual parameters, the socket immediately sends service requests to the underlying `Tcp` protocol module.

4.3.3 Setting up Connections

Since TCP is a connection-oriented protocol, a connection must be established before applications can exchange data. On one side, the application listens at a local address and port for incoming TCP connections:

```
socket.bind(Ipv4Address("10.0.0.42"), 42); // local address/port
socket.listen(); // start listening for incoming connections
```

On the other side, the application connects to a remote address and port to establish a new connection:

```
socket.connect(Ipv4Address("10.0.0.42"), 42); // remote address/port
```

4.3.4 Accepting Connections

The `Tcp` module automatically notifies the `TcpSocket` about incoming connections. The socket, in turn, notifies the application using the `ICallback::socketAvailable` method of the callback interface. Finally, incoming TCP connections must be accepted by the application before they can be used:

```
class TcpServerApp : public cSimpleModule, public TcpSocket::ICallback
{
    TcpSocket serverSocket; // server socket listening for connections
    SocketMap socketMap; // container for all accepted connections

    void socketAvailable(TcpSocket *socket, TcpAvailableInfo *info);
};

void TcpServerApp::socketAvailable(TcpSocket *socket, TcpAvailableInfo *info)
```

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```
{
    auto newSocket = new TcpSocket(info); // create socket using received info
    // ...
    socketMap.addSocket(newSocket); // store accepted connection
    serverSocket.accept(info->getNewSocketId()); // notify Tcp module
}
```

After the connection is accepted, the `Tcp` module notifies the application about the socket being established and ready to be used.

4.3.5 Sending Data

After the connection has been established, applications can send data to the remote application via a simple function call:

```
socket.send(packet1);
// ...
socket.send(packet42);
```

Packet data is enqueued by the local `Tcp` module and transmitted over time according to the protocol logic.

4.3.6 Receiving Data

The socket can be in one of two modes: “autoread” and “explicit-read”. In “autoread” mode, TCP immediately forwards received data to the socket, whereas in “explicit-read” mode, the application must explicitly request data by calling the `read()` method of `TcpSocket`. By default, the socket is created in “autoread” mode; “explicit-read” mode can be selected by calling `setAutoRead(false)` before the connection is established. Note that sockets created by forking off a listening socket inherit the mode from the listening socket.

Receiving data is done by implementing the corresponding method of the `TcpSocket::ICallback` interface. One caveat is that packet data may arrive in different chunk sizes (but the same order) than they were sent, due to the nature of TCP protocol.

For example, the application may directly implement the `TcpSocket::ICallback` interface and print the received data as follows:

```
class TcpApp : public cSimpleModule, public TcpSocket::ICallback
{
    void socketDataArrived(TcpSocket *socket, Packet *packet, bool urgent);
};

void TcpApp::socketDataArrived(TcpSocket *socket, Packet *packet, bool urgent)
{
    EV << packet->peekData() << endl;
}
```

4.4 SCTP Socket

The `SctpSocket` class provides an easy to use C++ interface to manage SCTP connections, and to send and receive data. The underlying SCTP protocol is implemented in the `Sctp` module.

4.4.1 Callback Interface

Messages received from the `Sctp` module can be processed by the `SctpSocket` where they belong to. The `SctpSocket` deconstructs the message and uses the `SctpSocket::ICallback` interface to notify the application about the received data or service indication:

```
class ICallback // inner class of SctpSocket
{
    void socketDataArrived(SctpSocket* socket, Packet *packet, bool urgent);
    void socketEstablished(SctpSocket *socket);
    // ...
    void socketClosed(SctpSocket *socket);
    void socketFailure(SctpSocket *socket, int code);
};
```

4.4.2 Configuring Connections

The SctpSocket class supports setting several SCTP specific connection parameters directly:

```
socket.setOutboundStreams(2);
socket.setStreamPriority(1);
socket.setEnableHeartbeats(true);
// ...
```

Upon setting the individual parameters, the socket immediately sends service requests to the underlying Sctp protocol module.

4.4.3 Setting up Connections

Since SCTP is a connection-oriented protocol, a connection must be established before applications can exchange data. On one side, the application listens at a local address and port for incoming SCTP connections:

```
socket.bind(Ipv4Address("10.0.0.42"), 42); // local address/port
socket.listen(true); // start listening for incoming connections
```

On the other side, the application connects to a remote address and port to establish a new connection:

```
socket.connect(Ipv4Address("10.0.0.42"), 42);
```

4.4.4 Accepting Connections

The Sctp module automatically notifies the SctpSocket about incoming connections. The socket, in turn, notifies the application using the ICallback::socketAvailable method of the callback interface. Finally, incoming SCTP connections must be accepted by the application before they can be used:

```
class SctpServerApp : public cSimpleModule, public SctpSocket::ICallback
{
    SocketMap socketMap;
    SctpSocket serverSocket;

    void socketAvailable(SctpSocket *socket, SctpAvailableInfo *info);
};

void SctpServerApp::socketAvailable(SctpSocket *socket, SctpAvailableInfo *info)
{
    auto newSocket = new SctpSocket(info);
    // ...
    socketMap.addSocket(newSocket);
    serverSocket.accept(info->getNewSocketId());
}
```

4.4.5 Sending Data

After the connection has been established, applications can send data to the remote application via a simple function call:

```
socket.send(packet1);  
// ...  
socket.send(packet42);
```

Packet data is enqueued by the local `Sctp` module and transmitted over time according to the protocol logic.

4.4.6 Receiving Data

Receiving data is as simple as implementing the corresponding method of the `SctpSocket::ICallback` interface. One caveat is that packet data may arrive in different chunk sizes (but the same order) than they were sent due to the nature of SCTP protocol.

For example, the application may directly implement the `SctpSocket::ICallback` interface and print the received data as follows:

```
class SctpApp : public cSimpleModule, public SctpSocket::ICallback  
{  
    void socketDataArrived(SctpSocket *socket, Packet *packet, bool urgent);  
};  
  
void SctpApp::socketDataArrived(SctpSocket *socket, Packet *packet, bool urgent)  
{  
    EV << packet->peekData() << endl;  
}
```

4.5 IPv4 Socket

The `Ipv4Socket` class provides an easy to use C++ interface to send and receive IPv4 datagrams. The underlying IPv4 protocol is implemented in the `Ipv4` module.

4.5.1 Callback Interface

Messages received from the `Ipv4` module must be processed by the socket where they belong as shown in the general section. The `Ipv4Socket` deconstructs the message and uses the `Ipv4Socket::ICallback` interface to notify the application about the received data:

```
class ICallback // inner class of Ipv4Socket  
{  
    void socketDataArrived(Ipv4Socket *socket, Packet *packet);  
};
```

4.5.2 Configuring Sockets

In order to only receive IPv4 datagrams sent to a specific local address or containing a specific protocol, the socket can be bound to the desired local address or protocol.

For example, the following code fragment shows how the INET `PingApp` binds to the ICMPv4 protocol to receive all incoming ICMPv4 Echo Reply messages:

```
socket.bind(&Protocol::icmpv4, Ipv4Address()); // filter for ICMPv4 messages
```

For only receiving IPv4 datagrams from a specific remote address, the socket can be connected to the desired remote address:

```
socket.connect(Ipv4Address("10.0.0.42")); // filter for remote address
```

4.5.3 Sending Data

After the socket has been configured, applications can immediately start sending IPv4 datagrams to a remote address via a simple function call:

```
socket.sendTo(packet, Ipv4Address("10.0.0.42")); // remote address
```

If the application wants to send several IPv4 datagrams to the same destination address, it can optionally connect to the destination:

```
socket.connect(Ipv4Address("10.0.0.42")); // remote address
socket.send(packet1);
// ...
socket.send(packet42);
```

The IPv4 protocol is in fact connectionless, so when the `Ipv4` module receives the connect request, it simply remembers the remote address and uses it as the default destination address for later sends.

The application can call `connect()` several times on the same socket.

4.5.4 Receiving Data

For example, the application may directly implement the `Ipv4Socket::ICallback` interface and print the received data as follows:

```
class Ipv4App : public cSimpleModule, public Ipv4Socket::ICallback
{
    void socketDataArrived(Ipv4Socket *socket, Packet *packet);
};

void Ipv4App::socketDataArrived(Ipv4Socket *socket, Packet *packet)
{
    EV << packet->peekData() << endl;
}
```

4.6 IPv6 Socket

The `Ipv6Socket` class provides an easy to use C++ interface to send and receive IPv6 datagrams. The underlying IPv6 protocol is implemented in the `Ipv6` module.

4.6.1 Callback Interface

Messages received from the `Ipv6` module must be processed by the socket where they belong as shown in the general section. The `Ipv6Socket` deconstructs the message and uses the `Ipv6Socket::ICallback` interface to notify the application about the received data:

```
class ICallback // inner class of Ipv6Socket
{
    void socketDataArrived(Ipv6Socket *socket, Packet *packet);
};
```

4.6.2 Configuring Sockets

In order to only receive IPv6 datagrams sent to a specific local address or containing a specific protocol, the socket can be bound to the desired local address or protocol.

For example, the following code fragment shows how the INET `PingApp` binds to the ICMPv6 protocol to receive all incoming ICMPv6 Echo Reply messages:

```
socket.bind(&Protocol::icmpv6, Ipv6Address()); // filter for ICMPv6 messages
```

For only receiving IPv6 datagrams from a specific remote address, the socket can be connected to the desired remote address:

```
socket.connect(Ipv6Address("10:0:0:0:0:0:0:42")); // filter for remote address
```

4.6.3 Sending Data

After the socket has been configured, applications can immediately start sending IPv6 datagrams to a remote address via a simple function call:

```
socket.sendTo(packet, Ipv6Address("10:0:0:0:0:0:0:42")); // remote address
```

If the application wants to send several IPv6 datagrams to the same destination address, it can optionally connect to the destination:

```
socket.connect(Ipv6Address("10:0:0:0:0:0:0:42")); // remote address
socket.send(packet1);
// ...
socket.send(packet42);
```

The IPv6 protocol is, in fact, connectionless, so when the `Ipv6` module receives the connect request, it simply remembers the remote address and uses it as the default destination for later sends.

The application can call `connect()` several times on the same socket.

4.6.4 Receiving Data

For example, the application may directly implement the `Ipv6Socket::ICallback` interface and print the received data as follows:

```
class Ipv6App : public cSimpleModule, public Ipv6Socket::ICallback
{
    void socketDataArrived(Ipv6Socket *socket, Packet *packet);
};

void Ipv6App::socketDataArrived(Ipv6Socket *socket, Packet *packet)
{
    EV << packet->peekData() << endl;
}
```

4.7 L3 Socket

The `L3Socket` class provides an easy to use C++ interface to send and receive datagrams using the conceptual network protocols. The underlying network protocols are implemented in the `NextHopForwarding`, `Flooding`, `ProbabilisticBroadcast`, and `AdaptiveProbabilisticBroadcast` modules.

4.7.1 Callback Interface

Messages received from the network protocol module must be processed by the associated socket as shown in the general section. The `L3Socket` deconstructs the message and uses the `L3Socket::ICallback` interface to notify the application about the received data:

```
class ICallback // inner class of L3Socket
{
    void socketDataArrived(L3Socket *socket, Packet *packet);
};
```

4.7.2 Configuring Sockets

Since the `L3Socket` class is network protocol agnostic, it must be configured to connect to a desired network protocol:

```
L3Socket socket(&Protocol::flooding);
```

In order to only receive datagrams which are sent to a specific local address or contain a specific protocol, the socket can be bound to the desired local address or protocol. The conceptual network protocols can work with the `ModuleIdAddress` class which contains a `moduleId` of the desired network interface.

For example, the following code fragment shows how the INET `PingApp` binds to the Echo protocol to receive all incoming Echo Reply messages:

```
socket.bind(&Protocol::echo, ModuleIdAddress(42));
```

For only receiving datagrams from a specific remote address, the socket can be connected to the desired remote address:

```
socket.connect(ModuleIdAddress(42)); // filter for remote interface
```

4.7.3 Sending Data

After the socket has been configured, applications can immediately start sending datagrams to a remote address via a simple function call:

```
socket.sendTo(packet, ModuleIdAddress(42)); // remote interface
```

If the application wants to send several datagrams to the same destination address, it can optionally connect to the destination:

```
socket.connect(ModuleIdAddress(42)); // remote interface
socket.send(packet1);
//..
socket.send(packet42);
```

The network protocols are, in fact, connectionless, so when the protocol module receives the connect request, it simply remembers the remote address, and uses it as the default destination address for later sends.

The application can call `connect()` several times on the same socket.

4.7.4 Receiving Data

For example, the application may directly implement the `L3Socket::ICallback` interface and print the received data as follows:

```
class L3App : public cSimpleModule, public L3Socket::ICallback
{
    void socketDataArrived(L3Socket *socket, Packet *packet);
```

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```
};

void L3App::socketDataArrived(L3Socket *socket, Packet *packet)
{
    EV << packet->peekData() << endl;
}
```

4.8 TUN Socket

The `TunSocket` class provides an easy to use C++ interface to send and receive datagrams using a TUN interface. The underlying TUN interface is implemented in the `Tun` module.

A TUN interface is basically a virtual network interface which is usually connected to an application (from the outside) instead of other network devices. It can be used for many networking tasks such as tunneling, or virtual private networking.

4.8.1 Callback Interface

Messages received from the `Tun` module must be processed by the socket where they belong as shown in the general section. The `TunSocket` deconstructs the message and uses the `TunSocket::ICallback` interface to notify the application about the received data:

```
class ICallback // inner class of TunSocket
{
    void socketDataArrived(TunSocket *socket, Packet *packet);
};
```

4.8.2 Configuring Sockets

A `TunSocket` must be associated with a TUN interface before it can be used:

```
socket.open(interface->getId());
```

4.8.3 Sending Packets

As soon as the `TunSocket` is associated with a TUN interface, applications can immediately start sending datagrams via a simple function call:

```
socket.send(packet);
```

When the application sends a datagram to a `TunSocket`, the packet appears for the protocol stack within the network node as if the packet were received from the network.

4.8.4 Receiving Packets

Messages received from the TUN interface must be processed by the corresponding `TunSocket`. The `TunSocket` deconstructs the message and uses the `TunSocket::ICallback` interface to notify the application about the received data:

```
class TunApp : public cSimpleModule, public TunSocket::ICallback
{
    void socketDataArrived(TunSocket *socket, Packet *packet);
};

void TunApp::socketDataArrived(TunSocket *socket, Packet *packet)
{
```

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```
EV << packet->peekData() << endl;  
}
```

When the protocol stack within the network node sends a datagram to a TUN interface, the packet appears for the application which uses a `TunSocket` as if the packet were sent to the network.

TESTING

INET contains a comprehensive test suite. The test suite can be found under the `tests` folder. Testing is usually done from the command line after setting up the environment with:

```
$ cd ~/workspace/omnetpp
$ . setenv
$ cd ~/workspace/inet
$ . setenv
```

5.1 Regression Testing

The most often used tests are the so-called fingerprint tests, which are generally useful for regression testing. Fingerprint tests are a low-cost but effective tool for regression testing of simulation models during development. For more details on fingerprint testing, please refer to the [OMNeT++ manual](#).

The INET fingerprint tests can be found under the `tests/fingerprint` folder. Each test is basically a simulation scenario described by one line in a CSV file. INET contains several such CSV files for different groups of tests. One CSV line describes a simulation by specifying the working directory, command line arguments, simulation time limit, expected fingerprint and test result (e.g. pass or fail), and several user-defined tags to help filtering.

Fingerprint tests can be run using the **fingerprinttest** script found in the above folder. To run all fingerprint tests, simply run the script without any arguments:

```
$ ./fingerprinttest
```

The script also has various command line arguments, which can be understood by asking for help with:

```
$ ./fingerprinttest -h
```

Running all INET fingerprint tests takes several minutes even on a modern computer. By default, the script utilizes all available CPUs, the tests are run in parallel (non-deterministic order).

The simplest way to make the tests run faster is to run only a subset of all fingerprint tests. The script provides a filter parameter (`-m`) which takes a regular expression that is matched against all information of the test.

Another less commonly used technique is running the fingerprint tests in release mode. One disadvantage of release mode is that certain assertions, which are conditionally compiled, are not checked.

For example, the following command runs all wireless tests in release mode:

```
$ ./fingerprinttest --release -m wireless
```

While the tests are running, the script prints some information about each test followed by the test result. The test result is either PASS, FAIL, or ERROR (plus some optional details):

```
/examples/bgpv4/BgpAndOspfSimple/ -f omnetpp.ini -c config1 -r 0 ... : PASS
/examples/bgpv4/BgpCompleteTest/ -f omnetpp.ini -c config1 -r 0 ... : FAIL
```

When testing is finished, the script also prints a short summary of the results with the total number of tests, failures, and errors:

```
Failures:
  /examples/bgpv4/BgpCompleteTest/ -f omnetpp.ini -c config1 -r 0
-----
Ran 2 tests in 3.615s

FAILED (failures=1)
```

Apart from the information printed on the console, fingerprint testing also produces three CSV files. One for the updated fingerprints, one for the failed tests, and one for the test which couldn't finish without an error. The updated file can be used to overwrite the original CSV file to accept the new fingerprints.

Usually, when a simulation model is being developed, the fingerprint tests should be run every now and then to make sure there are no regressions (i.e. all tests run with the expected result). How often tests should be run depends on the kind of development; tests may be run from several times a day to a few times a week.

For example, if a completely new simulation model is developed with new examples, then the new fingerprint tests can be run as late as when the first version is pushed into the repository. In contrast, during an extensive model refactoring, it's advisable to run the affected tests more often (e.g. after each small step). Running the tests more often helps avoid getting into a situation where it's difficult to tell if the new fingerprints are acceptable or not.

In any case, certain correct changes in the simulation model break fingerprint tests. When this happens, there are two ways to validate the changes:

- One is when the changes are further divided until the smallest set of changes is found, which still break fingerprints. In this case, this set is carefully reviewed (potentially by multiple people), and if it is found to be correct, then the fingerprint changes are simply accepted.
- The other is to actually change the fingerprint calculation in a way that ignores the effects of the set of changes. To this end, you can change the fingerprint ingredients, use filtering, or change the fingerprint calculation algorithm in C++. These options are covered in the OMNeT++ manual.

With this method, the fingerprint tests must be re-run before the changes are applied using the modified fingerprint calculation. The updated fingerprint CSV files must be used to run the fingerprint tests after applying the changes. This method is often time-consuming but may be worth the effort for complex changes because it allows having far greater confidence in correctness.

For a simple example, a timing change that neither changes the order of events nor where the events happen can be easily validated by changing the fingerprint and removing time from the ingredients (default is tplx) as follows:

```
./fingerprinttest -a --fingerprint=0 --fingerprint-ingredients=plx
```

Another common example is when a submodule is renamed. Since the default fingerprint ingredients contain the full module path of all events, this change will break fingerprint tests. If the module is not optional or not accessed by other modules using its name, then removing the module path from the fingerprint ingredients (i.e., using tlx) can be used to validate this change.

For a more complicated example, when the IEEE 802.11 MAC (very complex) model is refactored to a large extent, then it's really difficult to validate the changes. If despite the changes, the model is expected to keep its external behavior, then running the fingerprint tests can be used to prove this to some extent. The easy way to do this is to actually ignore all events executed by the old and the new IEEE 802.11 MAC models:

```
# please note the quoting uses both ' and " in the right order
./fingerprinttest -a --fingerprint-modules="'not(fullPath(**.wlan[*].mac.**))'"
```

INET uses continuous integration on Github. Fingerprint and other tests are run automatically for changes on the master and integration branches. Moreover, all submitted pull requests are automatically tested the same way. The result of the test suite is clearly marked on the pull request with a check mark or a red cross.

In general, contributors are expected to take care of not breaking the fingerprint tests. In the case of a necessary fingerprint change, the CSV files should be updated in separate patches.

5.1.1 INET Fingerprint Ingredients

In addition to the standard OMNeT++ fingerprint ingredients (covered in the [OMNeT++ manual](#)), INET provides 5 additional fingerprint ingredients that offer additional control over what aspects of network simulation are considered significant for reproducibility testing.

These ingredients are handled by the `inet::FingerprintCalculator` class. The fingerprint test script automatically adds the corresponding `--fingerprint-calculator-class` option to the simulation command lines.

Filter Ingredients (Exclusion Filters)

These ingredients act as filters that remove certain events from fingerprint calculation entirely, before other ingredients are processed:

~ (NETWORK_COMMUNICATION_FILTER)

Excludes intra-node communication from fingerprint calculation. When enabled, only includes packet arrival events that cross network node boundaries (where sender node and arrival node are different); all other events are excluded from the fingerprint. This focuses fingerprinting on actual network communication rather than internal node processing.

U (PACKET_UPDATE_FILTER)

Excludes packet update arrival events from fingerprint calculation, all other events are unaffected. When enabled, filters out packets that are marked as “transmission update” packets. This causes packet transmission updates such as truncations to be ignored. Packets will only be taken into account with their originally intended content, regardless of what was actually transmitted. This flag is formally independent of the ‘~’ flag, but in practice it only makes sense together with it.

Path Information Ingredients

These ingredients add path information to the fingerprint hash for packet arrival events:

N (NETWORK_NODE_PATH)

For packet arrival events, includes the full path of both sender and arrival network nodes in the fingerprint. Ignores non-packet events and all events that happen outside network nodes. Makes the fingerprint sensitive to which specific network nodes are involved in communication. For each packet arrival event, adds the full path of the containing network node for both the sender and arrival modules, even inside the same network node. This is mostly useful when used together with the ~ ingredient.

I (NETWORK_INTERFACE_PATH)

For packet arrival events, includes the full interface path of both sender and arrival network interfaces in the fingerprint. Ignores non-packet events and all events that happen outside network interfaces. Makes the fingerprint sensitive to which specific network interfaces are used for communication. For each packet arrival event, adds the interface full path of the containing NIC modules, even inside the same network interface. This is mostly useful when used together with the ~ ingredient.

Content Ingredient

D (PACKET_DATA)

Includes the actual packet data in the fingerprint. For packet arrival events with non-zero length, adds the raw packet content to the hash. This makes the fingerprint sensitive to actual packet content, not just packet structure or timing. Note that this ingredient requires packet serialization, so it will not work for simulations that contain protocol headers, payload, etc. that have no serializer implemented.

Usage Notes

- Filter ingredients (~ and U) work as exclusion filters and are processed before other ingredients
- Path and content ingredients (N, I, D) only apply to packet arrival events

APPENDIX: AUTHOR'S GUIDE

6.1 Overview

This chapter is intended for authors and contributors of this *INET Developer's Guide*, and covers the guidelines for deciding what type of content is appropriate for this *Guide* and what is not.

The main guiding principle is to avoid redundancy and duplication of information with other pieces of documentation, namely:

- Standards documents (RFCs, IEEE specifications, etc.) that describe protocols that INET modules implement;
- *INET User's Guide*, which is intended for users who are interested in assembling simulations using the components provided by INET;
- *INET Framework Reference*, directly generated from NED and MSG comments by OMNeT++ documentation generator;
- Showcases, tutorials and simulation examples (`showcases/`, `tutorials/` and `examples/` folders in the INET project)

Why is duplication to be avoided? Multiple reasons:

- It is a waste of our reader's time if they have to skip information they have already seen elsewhere.
- The text can easily get out of date as the INET Framework evolves.
- It is extra effort for maintainers to keep all copies up to date.

6.2 Guidelines

6.2.1 Do Not Repeat the Standard

When describing a module that implements protocol X, do not go into lengths explaining what protocol X does and how it works, because that is appropriately (and usually, much better) explained in the specification or books on protocol X. It is OK to summarize the protocol's goal and principles in a short paragraph, though.

In particular, do not describe the *format of the protocol messages*. It surely looks nice and takes up a lot of space, but the same information can probably be found in myriad places all over the internet.

6.2.2 Do Not Repeat NED Documentation

Things like module parameters, gate names, emitted signals, and collected statistics are appropriately and formally part of the NED definitions, and there is no need to duplicate that information in this *Guide*.

Detailed information on the module, such as *usage details* and the list of *implemented standards*, should be covered in the module's NED documentation, not in this *Guide*.

6.2.3 Do Not Repeat C++ Documentation

Describing every minute detail of C++ classes, methods, arguments are expected to be appropriately present in their *doxygen* documentation.

6.2.4 What then?

Concentrate on giving a “big picture” of the implementation: what it is generally capable of, how the parts fit together, how to use the provided APIs, what were the main design decisions, etc. Give simple yet meaningful examples and just enough information about the API that after a quick read, users can “bootstrap” into implementing their own protocols and applications. If they have questions afterward, they will/should refer to the C++ documentation.